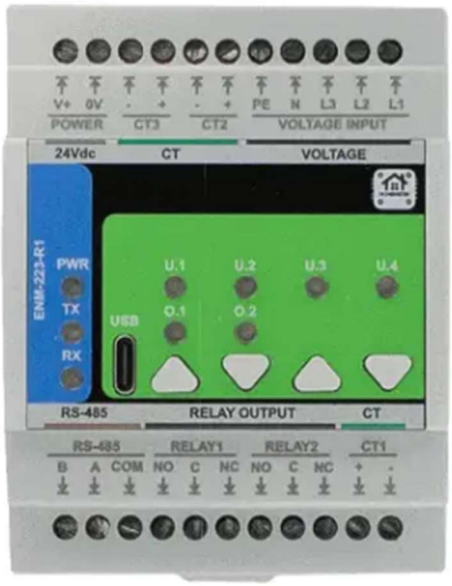
# Datasheet

## ENM-223-R1 Module

**Part No. ENM-223-R1**




**Manufacturer:** ISYSTEMS AUTOMATION S.R.L.
**Hardware Version:** V1.0
**Architecture:** Modular (MCU Board + Relay Board + Field Board)
**Controller:** RP2350

## Functional Overview

The system is implemented using three hardware boards:

| Board | Function |
|---|---|
| MCU Board | CPU, RS-485, USB-C, power conversion (24 V → 5 V → 3.3 V) |
| Field Board | 3-phase metering front-end (voltage dividers + CT inputs), isolated analog domain, field connectors |
| Relay Extension | 2 × SPDT relay output stage |

The ENM-223-R1 is a modular 3-phase energy metering + I/O module for power monitoring and automation. It measures 3 voltage channels (L1/L2/L3-N) and 3 current channels (external CTs) using a dedicated ATM90E32AS metering IC, and provides 2 SPDT relay outputs, plus optional 4 buttons / 4 user LEDs for local control and indication.

The module connects to a PLC or SCADA as a Modbus RTU slave over RS-485 and is configured via USB-C Web Serial (WebConfig).

## Technical Specifications

| Specification | Details |
|---|---|
| Microcontroller | RP2350A dual-core microcontroller |
| Storage | External QSPI Flash (W25Q32JV) |
| Power Input | 24 V DC nominal (V+ / 0V) |
| Metering IC | ATM90E32AS (3-phase energy metering / power quality) |
| Voltage Inputs | 3 × L1/L2/L3-N+PE sensing, 85–265 V AC (mains measurement inputs) |
| Current Inputs | 3 × CT channels (external CTs), 333 mV / 1 V RMS sensor inputs |
| Relay Outputs | 3 × SPDT — 3 A MAX continuous per output (board/system limit).<br>Relay component contacts rated up to 12 A @ 250 V AC (resistive). |
| User Interface | 4 buttons, 9xLEDs(Power, 4 user LEDs, RX, TX, 2xRelays) |
| RS-485 | RS-485 Modbus RTU (MAX485, half-duplex) with TVS surge protection, PTC fuses, EMI choke, and fail-safe biasing |
| USB | USB-C (ESD protected, configuration and firmware upload) |
| File System | LittleFS persistent configuration storage |

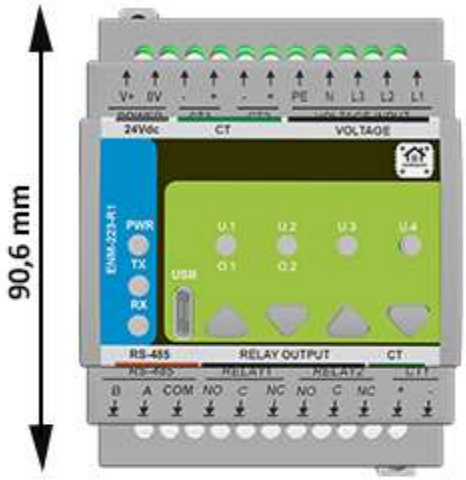
Note: Relay outputs are not internally fused. Use external overcurrent protection per local code and use an external contactor/relay for loads above 3 A or with high inrush/inductive characteristics.

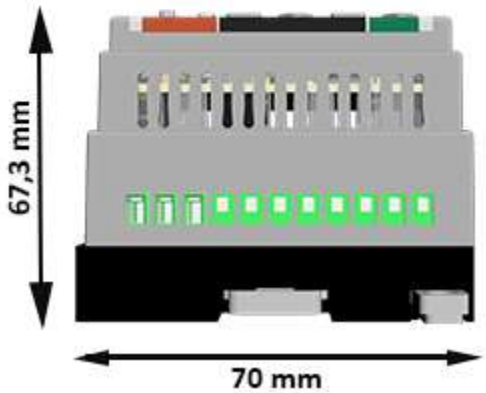
## Installation, Environmental & Mechanical

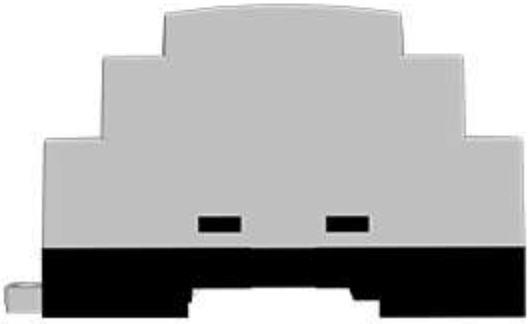
| Category | Specification | Details |
|---|---|---|
| Terminal Specifications | Terminal type | Pluggable screw terminal blocks, 5.08 mm pitch |
| | Terminal pitch | 5.08 mm |
| | Wire cross-section | 0.2–2.5 mm² (AWG 24–12) |
| | Conductor type | Solid or stranded copper |
| | Stranded wire | Ferrules recommended |
| | Tightening torque | 0.4 Nm (max) |
| Environmental Ratings | Operating temperature | 0 °C to +40 °C |
| | Storage temperature | -10 °C to 55 °C |
| | Relative humidity | 0–90 % RH, non-condensing |
| | Ingress protection | IP20 (inside cabinet) |
| | Installation | Indoor control cabinet only; not for outdoor or exposed installation |
| | Terminal protection | All wiring terminals must be protected against accidental contact |
| Mechanical & Packaging | Product dimensions | 71.5 x 90 x 59 mm (LxWxH) |
| | DIN units | 4 division units (≈ 72 mm DIN rail mounting width) |
| | Mounting | 35 mm DIN rail |
| | Net weight | Xxx g |
| | Gross weight | Xxx+98 g |
| | Pack size | 140× 125 × 94 mm (L × W × H) |
| | Mechanical drawing | Front + side views with DIN-clip depth (see below) |
| | Notes | All dimensions shown in millimeters unless stated otherwise |

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

**All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.**



## Cable Recommendations & Shield Grounding

Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

***General Routing Rules***

- Route low-level signal cables separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

***RS-485 Cable***

- **Construction:** twisted pair for **A/B**
- **Characteristic impedance: 120 Ω** (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples: J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**
- Use one twisted pair for **A/B**. Use the second pair for **COM (0V reference)** or spare.

***Shield Grounding***

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

## Compliance & Certifications

The HomeMaster MiniPLC is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

**EU Directives**

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

**Harmonised Standards**

| Area | Standards |
|---|---|
| EMC | EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions) |
| Electrical Safety | EN 62368-1 |
| RoHS | EN IEC 63000 |

**Radio**

The product integrates a pre-certified ESP32 Wi-Fi radio module (2.4 GHz). Final product conformity with the Radio Equipment Directive is demonstrated by the maintained technical documentation and conformity assessment of the complete device.

**Safety notice**

L / N: hazardous voltage · 24 V DC: SELV · Qualified personnel only

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